

Problem Set – Introduction to Functions.

1. Allow the user to repeatedly enter a quantity and price. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute the total (quantity times price). The function should be passed the quantity and price and then return the total. In the function, provide a 10% discount if the total is over \$10,0000.00. Display quantity, price and total. Sum and display the extended price.

INPUT	PROCESS	OUTPUT
QNT	comptotalprice(qnt,price) totprice= qnt * price If totprice > 10000 Discamt= totprice * 0.10 Else Discamt= 0 extprice = totprice - discamt return extprice	PRINT qnt
PRICE	Main extprice= 0 Do you want to do this program? (Y/N) While Y Input qnt, price Totprice = comptotalprice(qnt,price) Display totprice extprice= extprice + totprice Do you want to continue this program?	PRINT price
		PRINT extprice
		PRINT totprice

2. Enter players last name, number of hits and at bats at the keyboard. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average. Display last name and batting average. Give a count of the number of players entered.

INPUT	PROCESS	OUTPUT
LANAM	compbattavg(atbats, nuofhits) batavg= nuofhits/ atbats Return batavg	Print lanam

	player_count = 0	
NUOFHITS	Main Do you want to do this program? (y/n) While Y batavg = batavg+ nuofhits/ atbats player_count += 1 answer = input("Do you want to run the program again? (y/n): ") if player_count == 0: print("No information entered.")	Print batavg
ATBATS		Number of players entered

- Enter the destination city, miles travelled and gallons used for a trip. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute miles per gallon. Pass miles travelled and gallons used to the function. The function should return miles per gallon. Count the number of entries made (number of trips) Display destination city, miles and mpg. At end display the number of entries made.

INPUT	PROCESS	OUTPUT
DECITY	Comp milespergallon (mitrav, gaused) Milespergallon = mitrav/gaused Return milespergallon trip_count = 0	Print milespergallon Print trip_count Print decity
MITRAV	Main Do you want to do this program? (Y/N) While Y Milespergallon = mitrav/gaused trip_count += 1 answer = input("Do you want to run the program again? (y/n): ") if trip_count == 0: print("No information entered.")	
GAUSED		

4. Allow the employee to enter last name, job code and hours worked. Prompt the user on whether they want to do the program (Yes or No). Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay. Use Job code L is \$25/hr, A is \$30/hr and J is \$50/hr for respective pay rates. Compute gross pay. Give time and a half for overtime. Display last name and gross pay. Sum and display total of all gross pay.

INPUT	PROCESS	OUTPUT
LANAM	grosspay(howork, jocode) grosspay= howork * jocode Return grosspay employee_count = 0	Print lanam Print Grosspay Print totpay
HOWORK	Main Do you want to do this program? (Y/N) While Y Prompt user to enter job code If A, jocode == 30 If L, jocode == 25 Else, jocode == 50 If howork > 40, compute extended pay extrahours= howork-40 Overtime = extrahours * jocode * 1.5 totpay= overtime + grosspay employee_count += 1 answer = input("Do you want to run the program again? (y/n): ") if trip_count == 0: print("No information entered.")	
JOCODE		

5. Allow the user to enter student last name, credit hours and district code. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute tuition owed. Charge In district (code of I) \$250 per credit hour. Out of district (code of O) is \$550 per credit hour. The function should receive credit hours and district code and return tuition owed. Display student name and tuition owed. Sum and display total of all tuition owed.

INPUT	PROCESS	OUTPUT
LANAM	tuitioncost(crhours, dicode) tuitioncost= crhours* dicode Return tuitioncost student_tuition_count = 0	Print lanam Print tuitioncost Print student_tuition_count
CRHOURS	Main Do you want to do this program? (Y/N) While Y Prompt user to enter Dicode If I, crhour== 250 If O, jocode == 550 Tuitioncost = crhours * dicode student_tuition_count += 1 answer = input("Do you want to run the program again? (y/n): ") if student_tuition_count == 0: print("No information entered.")	
DICODE		